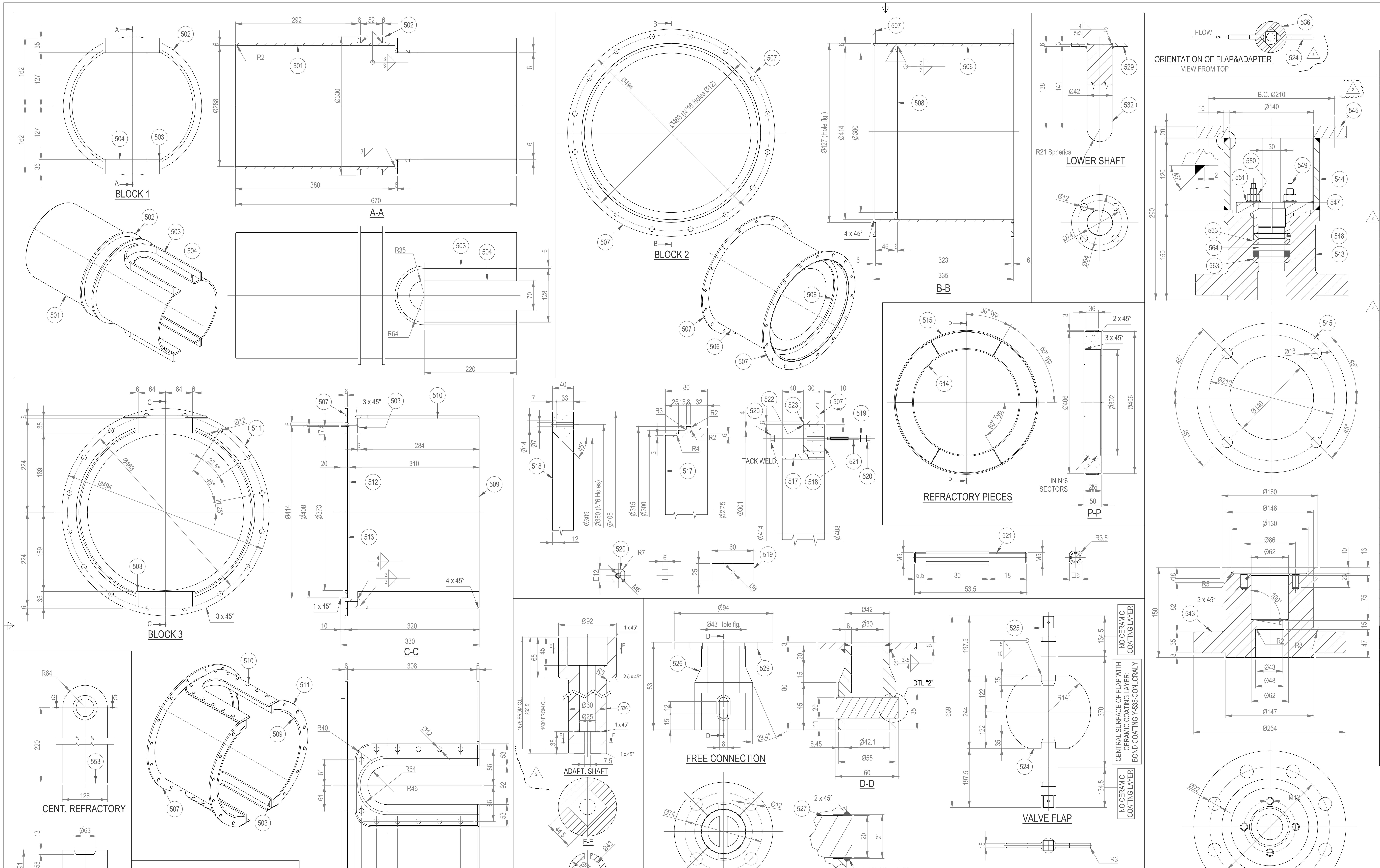

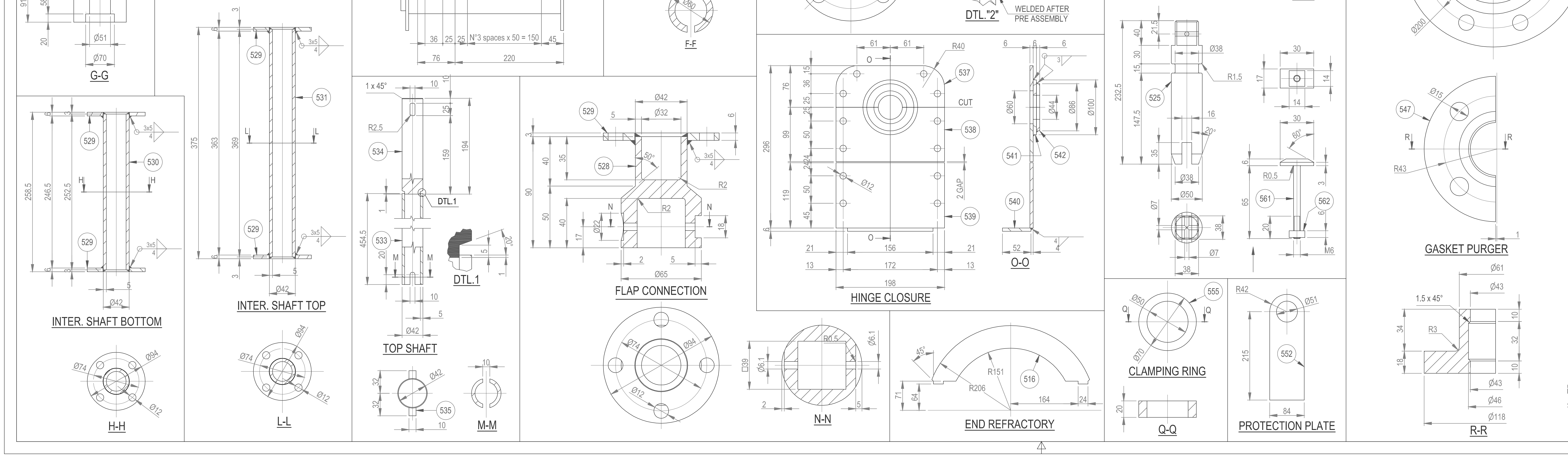

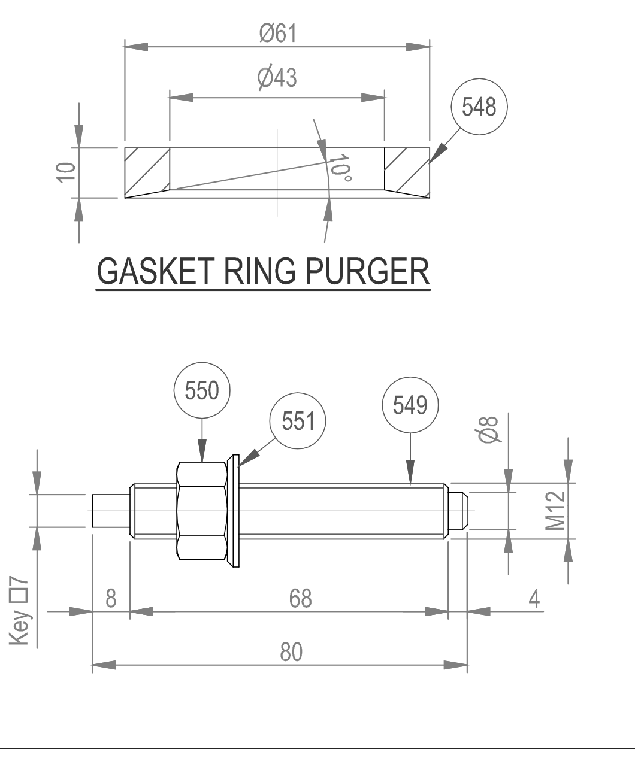

POS.	DESCRIZIONE/DESCRIPTION	DIMENSIONI/DIMENSIONS	MATERIALI/MATERIAL	Q.TY	CERTIFICATO CERTIFIED	SPARE PARTS	SPARE FOR TWO YEARS	NOTES
564	Guarnizione premistoppa/Baderna	Ø62/42 x Thk.10	KEMPCHEN TYPE K80 OR EQ.	2	2.2			
563	Guarnizione premistoppa/Baderna antiextrusion	Ø62/42 x Thk.10	KEMPCHEN TYPE K80S OR EQ.	2	2.2			
562	Dado quadro/Square nut	14 x 14 x Thk.6	SB-168 UNS N06025	2	3.1			
561	Spina antisfilamento/Pin for flap connection	See dwg.	SB-574 UNS N06025	2	3.1			
560	Guarnizione/Gasket K1	Ø4" 300Lb. RF	S.W. S.S. 321 + GRAPHITE	1	3.1			
559	Dado es./Hex. Nuts K1	Ø3/4"	SA-194 Gr.4	16	3.1			
558	Tiranti/Stud bolts K1	Ø3/4" x Lg.130	SA-193 Gr.B16	8	3.1			
557	Dado es./Hex. Nuts	M10	SB-564 UNS N06025	72	3.1			
556	Bullone/Bolts	M10 x Lg.25	SB-564 UNS N06025	72	3.1			
555	Guarnizione premistoppa/Stuffing gasket	Ø70/50 x Thk.20	GRAPHITE BRAIDED	2	2.2			
554	Refrattario/Refractory Block 3	Ø414/300 x Lg.289	CERAMIC FIBRE BLANKET	1	--			
553	Refrat. centraggio/Centering refractory	284 x 128 x Thk.91	STYRO-FOAM	2	--			
552	Lamiera protezione/Protection plate	257 x 84 x Thk.6	SB-168 UNS N06025	2	3.1			
551	Rondella/Washer	M12	C.S.	4	2.2			
550	Dado es./Hex. Nuts	M12	SA 194 Gr.2H	4	3.1			
549	Tiranti/Stud bolts	M12 see dwg.	SA 193 Gr.B7	4	3.1			
548	Anello premi guarn./Gasket ring purger	Ø61/43 x Thk.10	SA-387 Gr.11 Cl.2	1	3.1			
547	Premi guarnizione/Gasket purger	See dwg.	SA-387 Gr.11 Cl.2	2	3.1			
546								
545	Supporto attuatore/Actuator support	Ø250 x Thk.20	SA-387 Gr.11 Cl.2	1	3.1			
544	Piatto / Plate	#30 x Thk.10 Lg.120	SA-387 Gr.11 Cl.2	4	3.1			
543	Supporto flangiato/Flanged support	See dwg.	SA-336 Gr.F11 Cl.3	1	3.1			
542	Lamiera albero/Shaft plate	Ø86/44 x Thk.6	SB-168 UNS N06025	4	3.1			
541	Lamiera albero/Shaft plate	Ø100/44 x Thk.6	SB-168 UNS N06025	4	3.1			
540	Lamiera finale/Plate end	156 x 52 x Thk.6	SB-168 UNS N06025	2	3.1			
539	Lamiera di chiusura/Closed of plate	198 x 119 x Thk.6	SB-168 UNS N06025	2	3.1			
538	Lamiera di chiusura/Closed of plate	198 x 99 x Thk.6	SB-168 UNS N06025	2	3.1			
537	Lamiera di chiusura/Closed of plate	198 x 76 x Thk.6	SB-168 UNS N06025	2	3.1			
536	Adattatore albero/Shaft adapter	Øe92 x Lg.265.5	SA-105	1	3.1			
535	Linguetta/Key	25 x 10 x Thk.11	SB-168 UNS N06025	2	3.1			
534	Connessione attuatore/Actuator conn.	Øe42 x Lg.194	S.S. 321	1	2.2			
533	Albero superiore/Top shaft	Øe42 x Thk.5 Lg.454.5	SB-619 UNS N06059	1	3.1			
532	Tondo d'appoggio/Stop plug	Øe42 x Lg.162	SB-574 UNS N06025	1	3.1			
531	Albero interm./Interm. shaft top	Øe42 x Thk.5 Lg.369	SB-619 UNS N06059	1	3.1			
530	Albero interm./Interm. shaft bottom	Øe42 x Thk.5 Lg.252.5	SB-619 UNS N06059	1	3.1			
529	Flangia da lamiera/Plate of flange	Øe94 x Thk.6	SB-168 UNS N06025	8	3.1			
528	Connessione Flap/Flap connession	See dwg.	SB-574 UNS N06025	2	3.1			
527	Chiavetta/Key	20 x 60 x Thk.8	SB-168 UNS N06025	1	3.1			
526	Connessione libera/Free connection	See dwg.	SB-574 UNS N06025	1	3.1			
525	Perno valvola/Valve flap parte	Øe50 x Lg.232.5	SB-574 UNS N06025 + Ceramic coat	2	3.1			
524	Flap valvola/Valve flap	See dwg.	SB-168 UNS N06025 + Ceramic coat	1	3.1			
523	Anello/Ring	Øe426 x Thk.6 Lg.30	SB-168 UNS N06025	1	3.1			
522	Anello/Ring	Øe414 x Thk.3 Lg.40	SB-168 UNS N06025	1	3.1			
521	Tirante fissaggio/Refractory fixing	M5 x Lg.53.5 see dwg.	SB-574 UNS N06025	6	3.1			
520	Dado quadro/Square nut	M5 x Thk.6	SB-168 UNS N06025	12	3.1			
519	Piatto/Plate	60 x Thk.6 Lg.25	SB-168 UNS N06025	6	3.1			
518	Anello refrattario/Refractory ring	Ø408/309 x Thk.40	STYRO-FOAM	1	--			
517	Anello/Ring	Øe315 x Lg.80 see dwg.	SB-168 UNS N06025	1	3.1			
516	Refrattario finale/End refractory	Ø412/302 x Thk.25	STYRO-FOAM	2	--			
515	Anello blocca ref./Refractory ring	Øe412 x Thk.3 Lg.36	SB-168 UNS N06025	1	3.1			
514	Anello refrattario/Refractory ring	Ø406/302 x Thk.25	STYRO-FOAM	2	--			
513	Anello/Ring Block 3	Ø408/373 x Thk.6	SB-168 UNS N06025	1	3.1			
512	Virola /Barrel Block 3	Øe414 x Thk.3 Lg.20	SB-168 UNS N06025	1	3.1			
511	Piatto chiusura/Closure semiflange Block 3	Ø494/426 x Thk.6	SB-168 UNS N06025	2	3.1			
510	Piatto/Plate of hinge Block 3	296 x 198 x Thk.6	SB-168 UNS N06025	2	3.1			
509	Virola interna/Internal barrel Block 3	Øe426 x Thk.6 Lg.320	SB-168 UNS N06025	1	3.1			
508	Anello/Ring Block 2	Ø414/380 x Thk.6	SB-168 UNS N06025	1	3.1			
507	Flangia/Flange from plate Block 2/3	Ø494/427 x Thk.6	SB-168 UNS N06025	4	3.1			
506	Virola interna/Internal barrel Block 2	Øe426 x Thk.6 Lg.335	SB-168 UNS N06025	1	3.1			
505	Refrattario/Refractory Block 2	Ø414/300 x Lg.267	CERAMIC FIBRE BLANKET	1	--			
504	Piatto/Plate of hinge Block 1	284 x 128 x Thk.6	SB-168 UNS N06025	2	3.1			
503	Piatto/Plate bended Block 1-3	642 x 35 x Thk.6	SB-168 UNS N06025	4	3.1			
502	Anello/Ring Block 1	Ø330/300 x Thk.6	SB-168 UNS N06025	2	3.1			
501	Virola interna/Internal barrel Block 1	Øe300 x Thk.6 Lg.670	SB-168 UNS N06025	1	3.1			

3.2	Materiale collaudato da ente ufficiale. Material tested by official authority.	MATERIALS TEST SYMBOL (Certification Acc. TO EN-10204) Simbolismo collaudo materiali
3.1	Materiale con certificato del produttore. Material with manufacturer certificate.	
2.2	Materiale senza certificato del produttore. Material without manufacturer certificate.	

- Il materiale elencato in distinta è da intendersi valido per N°1 apparecchio.
  Quantity indicated on material list are for one shell.

LISTA MATERIALI / MATERIALS LIST

DISEGNO DEFINITIVO
Il cliente ha gia preso visione del disegno ed ha approvato le parti non evidenziate dall'ultima rev.
VALID FOR CONSTRUCTION
Drawing already approved by client except where indicated with last revision mark.
DATA: 10/06/2014 FIRMA: Sarti

OWNER : LORDEGAN PETROCHEMICAL COMPANY
PROJECT TITLE : LORDEGAN AMMONIA AND UREA PROJECT
CONTRACTOR : HAMPA ENGINEERING CORPORATION
DETAIL ENGINEER : HEDCO Hampa Energy Engineering & Design Company
P.O. NO. : 2103-POR-A-HM-255
ITEM NO. : 3-E-1401
REQ NO. : 2103-REQ-A-HM-255
DOC. NO. 2103-A-VD-255-A1-MD-028
Rev. 2

	NC – REVIEWED WITH NO COMMENT	DEPARTMENT :
	C – DO NOT PROCEED WITH FABRICATION	DATE:
	AC – APPROVED WITH MINOR COMMENTS	PERSON IN CHARGE :
	RE – DOCUMENT WAS NOT ACCEPTED	

Purchaser's permission to proceed or review taken on vendor prints shall not relieve vendor from its responsibilities or liabilities under purchase order.

Bypass insulation arrangement in general is a proprietaty solution specifically designed so to avoid carburization in the bypass pipe protrusion.
Same applies for the flap, which is basically operating at full temperature (ie, safe against metal dusting), with no request of a steam cooling system

REV.	DATE	DESCRIPTION	PREPARED	CHECKED	APPROVED
2	10/06/2014	Revised where indicated	Picenni	Sarti	Sarti
1	05/03/2014	Revised where indicated	Studical	Sarti	Sarti
0	29/04/2013	Issued for approval	Mazzoleni	Sarti	Sarti

ALFA LAVAL
SUISIO - BERGAMO - ITALY

CLIENTE CUSTOMER } Fimco
ORDINE CLIENTE PURCHASE ORDER } FZMC-13450-R1
IMPIANTO E LOC. LOCATION & UNIT } Ammonia Urea Project
SERVIZIO SERVICES } Sec. Reformer Effluent Waste Heat Boiler
TITOLO TITLE } Internal By-Pass & valve - details

SCALA/FORMATO SCALE/FORMAT	NO / A1
DATA DATE	29/04/2013
DISEGNATO DRAWN.	Mazzoleni
CONTROLL. CHECKED	Sarti
APPROVATO APPROVED	Sarti

TIPO / TEMA / TYPE: BXM SPEC.
NUMERO APPAR. NUMBER REQUIRED } 1
SIGLA / ITEM No.: 3-E-1401

DIMENSIONI / SIZE	NUM. DI FABBRICA SERIAL NUMBER	ANNO DI COS. YEAR OF CONS.	COMM. / JOB	NUM. DIS. / DWG. NUM.	REV.
13000 x 2025	12542	2013	7523	7523-01-500-02	2

HEDCO
NO COMMENT
By: seraj.mohammad
11:58 am, Jun 23, 2014

REV.	ISSUE DATE	DESCRIPTION	DWN. BY	CHKD. BY	APPR. BY	APPR. BY
			DISCIPLINE			PROJ.
2	10/06/14	REVISED WHERE INDICATED	SP	SS	SS	
1	05/03/14	REVISED WHERE INDICATED	AC	FG	SS	
0	29/04/13	ISSUED FOR APPROVAL	PM	FG	SS	

OWNER: LORDEGAN PETROCHEMICAL COMPANY
Project: LORDEGAN AMMONIA AND UREA PROJECT
CONTRACTOR : HAMPA ENGINEERING CORPORATION
DETAIL ENGINEER : HEDCO Hampa Energy Engineering & Design Company
ALFA LAVAL SUISIO - BERGAMO - ITALY

TITLE : INTERNAL BY-PASS & VALVE - DETAILS

Project No:2103
Scale:
Sheet of
HAMPA
Vendor
Doc. No. 7523-01-500-02
2103-A-VD-255-A1-MD-028
REV. 2

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